GNN MLP Transformer

(a)

MAPE (%)

300
100
30
10
3

6 7 8 9 10

Number of blocks

(b)

R^2

1 0 −2 −4 −6 −8 −10

Expanded view

1.0 0.5 0.0 −0.5

6 8 10

6 7 8 9 10

Number of blocks

(c) GNN

$R^2 = 0.85$
$MAPE = 11.16\%$
$RMSE = 44.53$
$MAE = 15.10$

6 blocks
7 blocks
8 blocks
9 blocks
10 blocks

Predicted χN^*

10^1 10^2 10^3

RPA χN^*

(d) MLP

$R^2 = -1.09$
$MAPE = 91.28\%$
$RMSE = 165.98$
$MAE = 64.70$

Predicted χN^*

10^1 10^2 10^3

RPA χN^*

(e) Transformer

$R^2 = 0.23$
$MAPE = 54.88\%$
$RMSE = 100.50$
$MAE = 49.77$

Predicted χN^*

10^1 10^2 10^3

RPA χN^*